

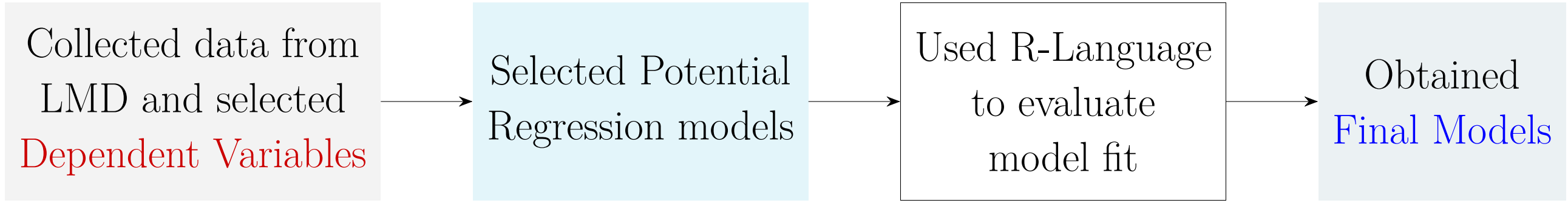
Abstract

Sudoku is a globally popular number puzzle game whose appeal lies in its simplicity and capacity to challenge logical and analytical skills, transcending cultural barriers and finding applications in various fields. The game is believed to bring many benefits to the players, including stimulating the mind, as well as developing their ability to make decisions (Uniyal, 2022). This research delves into evolving preferences by analyzing publicly available data from Logic Masters Deutschland (LMD), a leading German organization dedicated to expanding Sudoku logic puzzles through competitions and events. The dataset comprises a wealth of puzzle-related information, including puzzle names, author details (names, counts of solved and published puzzles across five difficulty levels), publication dates, solved puzzle counts, difficulty ratings, observations, comments, and the presence of puzzle variants like Killer, Arrows, Thermo, German, Reban, and Kropki Pairs. The data is limited to be from January 2023 to August 2023 to ensure it captures current trends while allowing enough time for general community engagement. Regression analyses of various kinds were employed, exploring the impact of various independent variables on the number of solved puzzles, puzzle difficulty rating, and puzzle satisfaction rating. These analyses unveil significant relationships among variables, yielding predictive models for future puzzle outcomes. By quantifying measurable puzzle’s success, the research sheds light on the intricate web of puzzle enthusiasts and their interaction with Sudoku and similar challenges.

Objectives

- Apply quantitative measures in the study of Sudoku variant puzzles.
- Employ regression analyses to investigate the influence of various factors on the puzzles’ number of solved, difficulty rating, and satisfaction rating
- Evaluate the ability of regression models to predict future puzzle outcomes.

Methods



Research Process

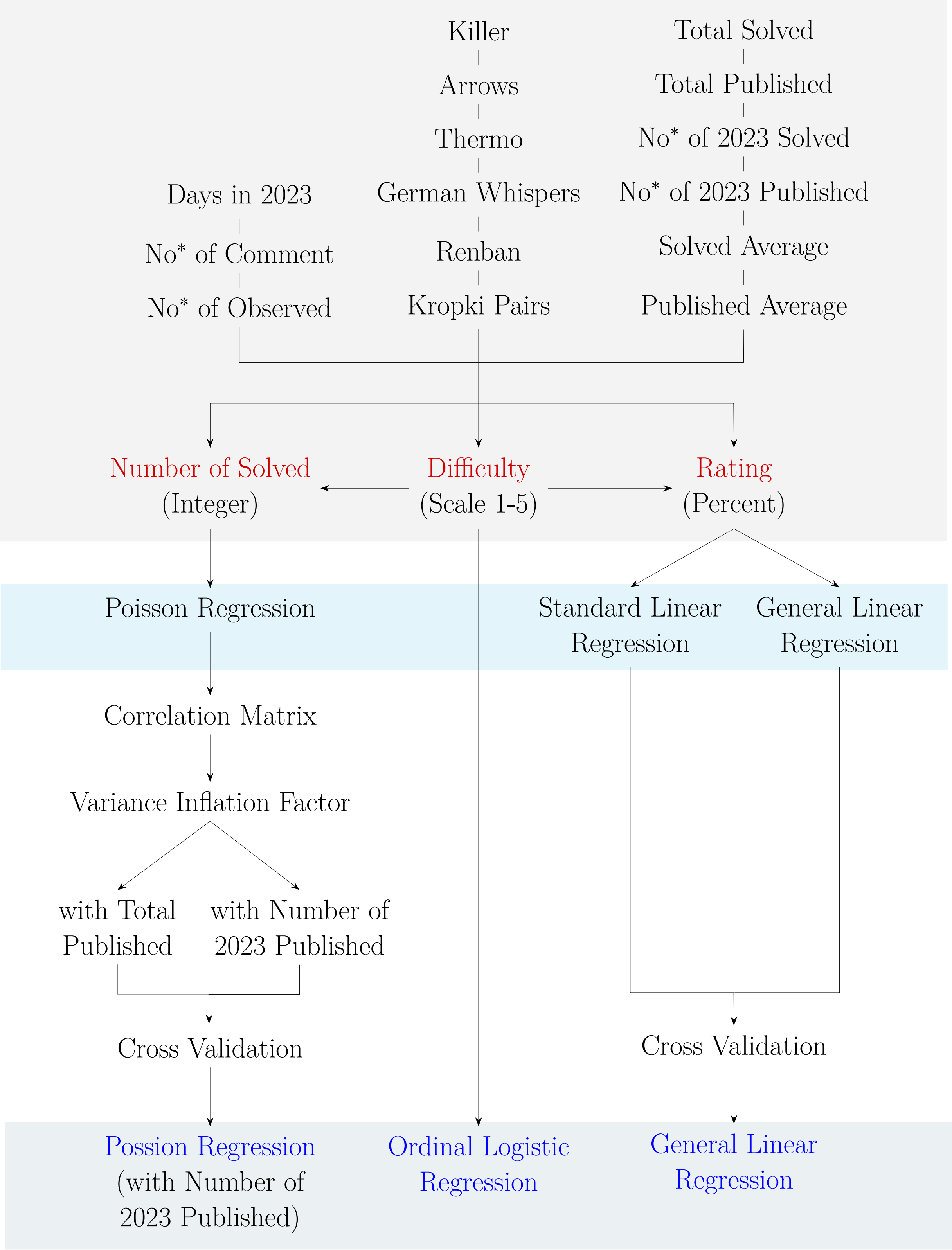


Figure 1. Research Summary

Result

I. Number of Solved - Poisson Regression

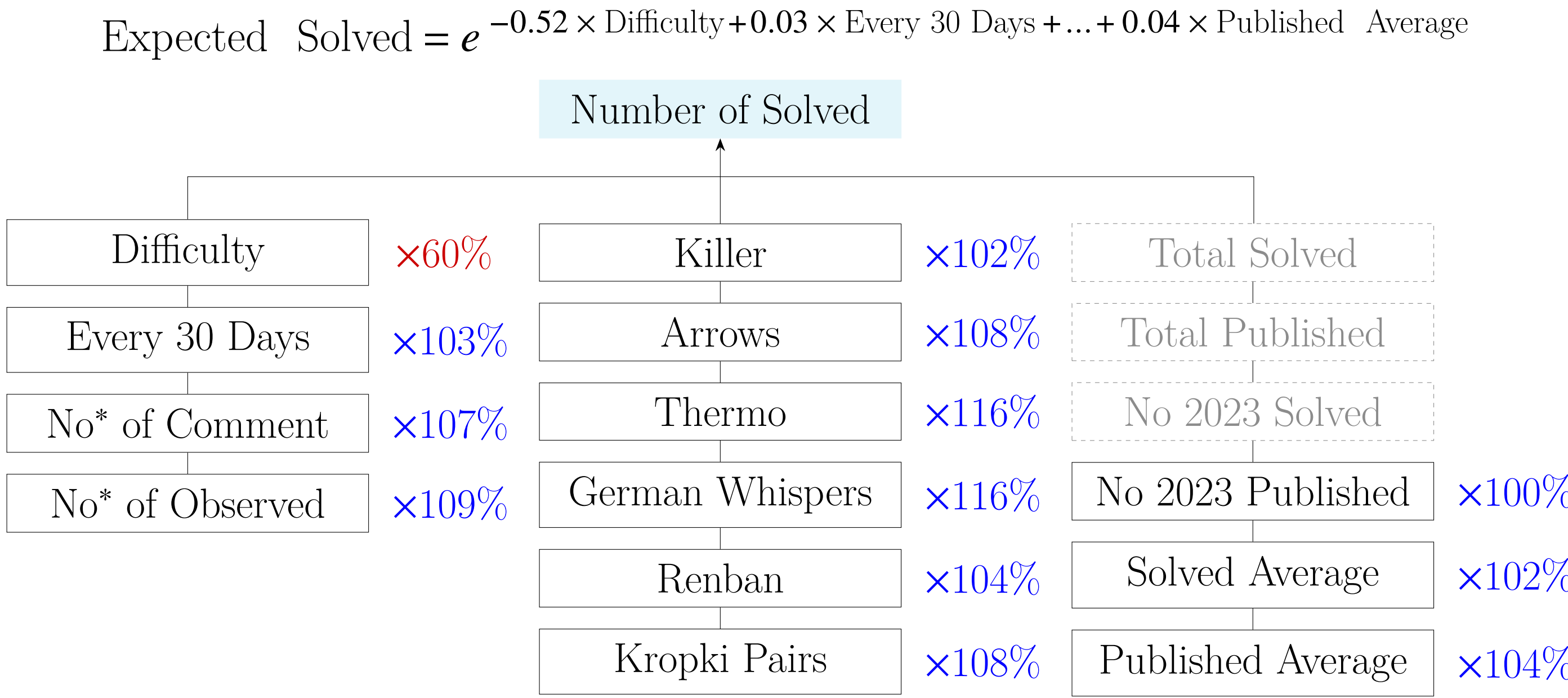


Figure 2. Number of Solved - Model Result

II. Rating - General Linear Regression

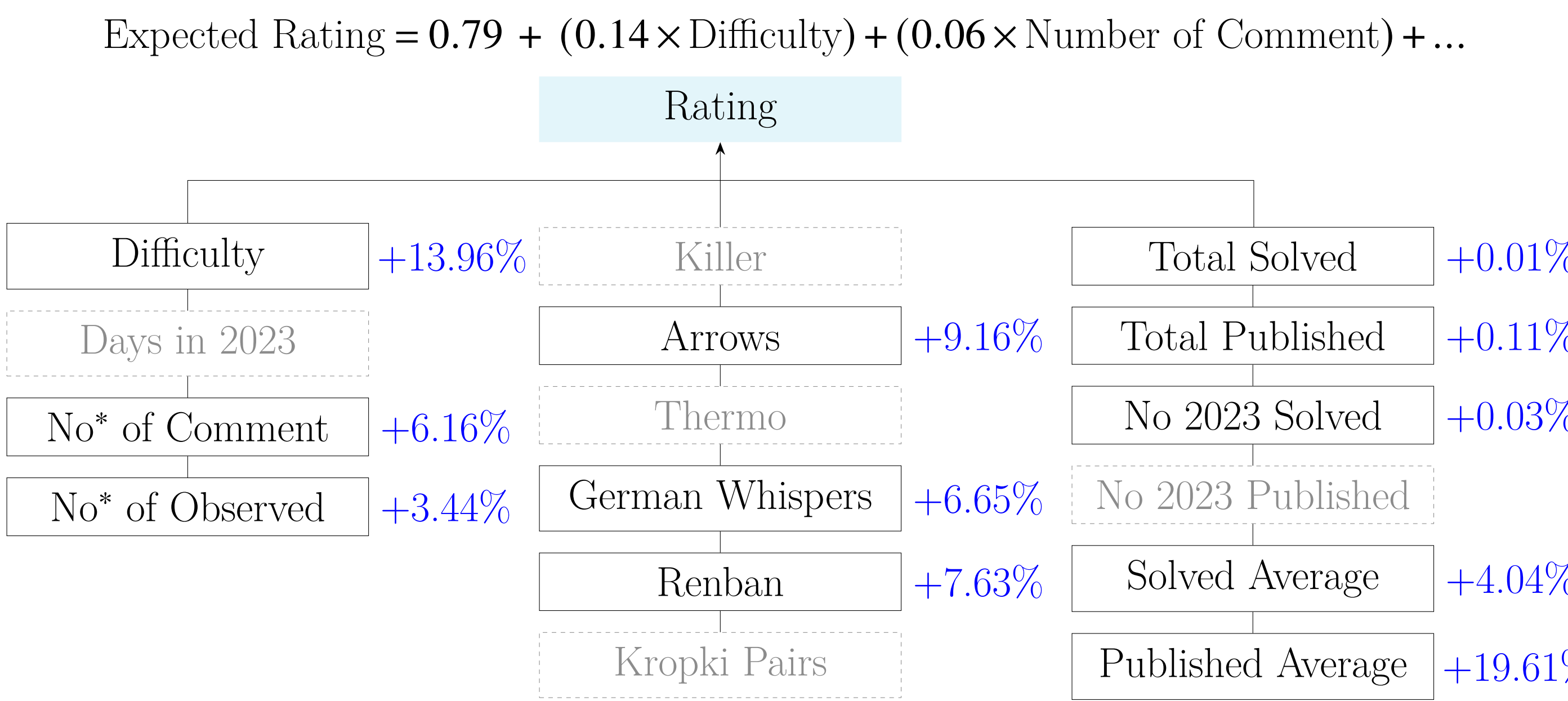


Figure 3. Rating - Model Result

III. Difficulty - Ordinal Logistic Regression

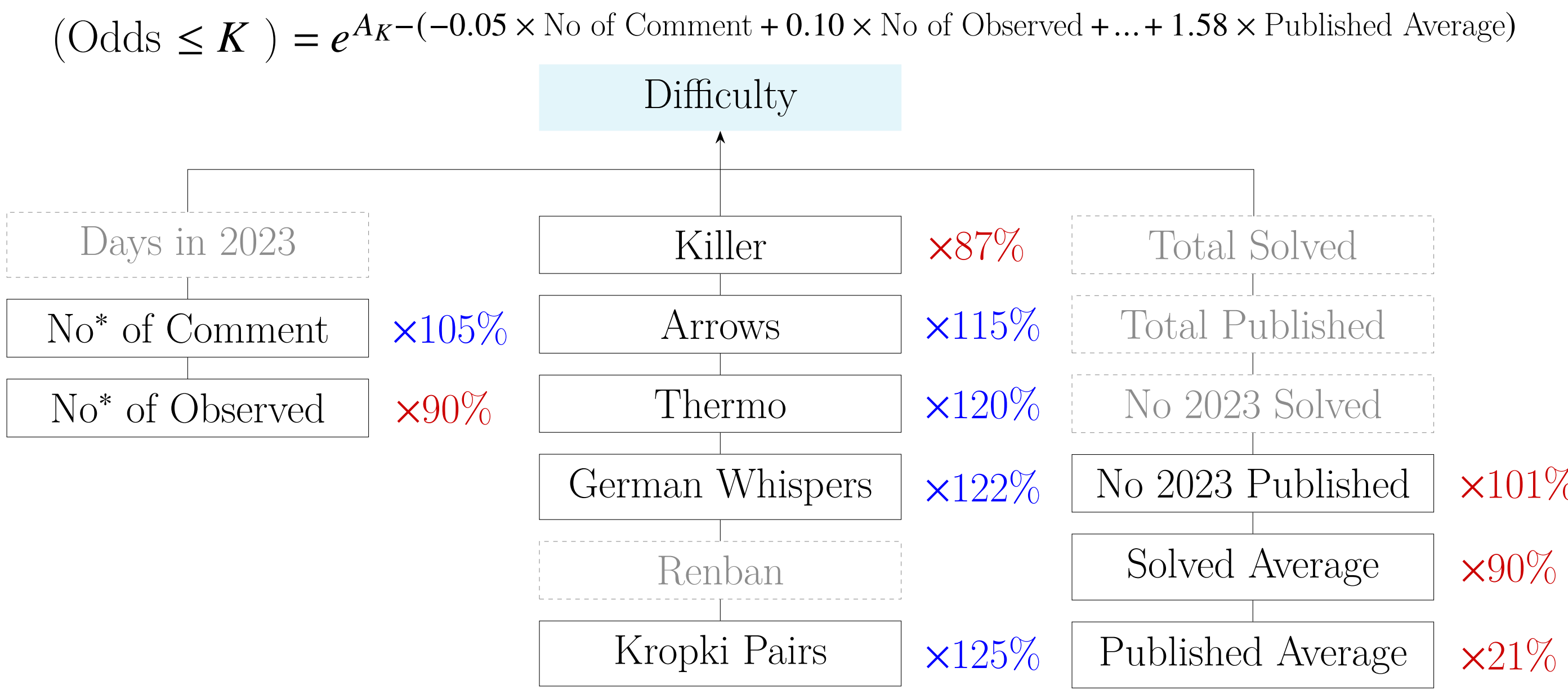


Figure 4. Difficulty - Model Result

Models Prediction Success Rate

September 2023 Prediction Success Rate

	True	False		True	False
Positive	87.3%	12.7%	Positive	62.4 %	37.6 %
Negative	79.7%	20.3%	Negative	78.3 %	21.7 %

Table 1. Number of Solved Success Rate

Table 2. Rating Success Rate

Difficulty Level within 1	88.2 %
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Table 3. Difficulty Success Rate

Discussion

- Quantitative methodology applied to game play can describe important aspects of the game itself
- New questions:
 - Are there any other unexplored factors?
 - Why does Killer variant contribute to the higher difficulty-level puzzle?
 - What are the factors that contribute to the decrease in puzzle rating?
- Further Study Suggestion:
 - Additional variables (i.e. variant types)
 - Longer time period
 - Different regression models

References

- Rätselportal — Logic masters Deutschland.
- Uniyal, P. (2022, September 9). International Sudoku day 2022: Health benefits of playing Sudoku every day. Hindustan Times.

Appendix

Coefficients of independent variables for each model:

	Number of Solved	Rating	Difficulty
Difficulty	-0.5232	0.1396	-
Days in 2023	0.0010	-	-
Number of Comment	0.0722	0.0616	-0.0527
Number of Observed	0.0880	0.0344	0.1008
Killer	0.0228	-	0.1442
Arrows	0.0800	0.0916	-0.1370
Thermo	0.1467	-	-0.1861
German Whispers	0.1462	0.0665	-0.2015
Renban	0.0449	0.0763	-
Kropki Pairs	0.0769	-	-0.2198
Total Solved	-	0.0001	-
Total Published	-	0.0011	-
Number of 2023 Solved	-	0.0003	-
Number of 2023 Published	0.0040	-	-0.0095
Solved Average	0.0244	0.0404	0.1095
Published Average	0.0414	0.1961	1.5835
A ₁	-	-	1.8380
A ₂	-	-	3.5904
A ₃	-	-	5.4224
A ₄	-	-	7.1852

Table 4. Coefficients of Independent Variables